Cost-Risk Regularization Timing in Portfolio Reinforcement Learning

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Cost-risk-aware portfolio reinforcement learning often activates cost and risk penalties from the first update. Using existing PPO results and market/model summaries, we examine when these penalties enter training. Under the same architecture, data, seeds, and friction proxy, Unreg, always-on Direct CR, and Adaptive Competence-before-Discipline (ACoDi) attain Sharpe 4.13, 3.79, and 4.09. ACoDi uses declining training turnover to switch persistently to fixed penalties; it is a scheduling heuristic, not a feasibility guarantee.

Conditional on valid source p-values, multiplicity checks retain the reported recovery over Direct CR, including a Bonferroni bound across all 21 main-table pairs, but establish neither superiority over Unreg nor cross-market generalization. Existing sweep, trigger, and cost-unit analyses distinguish observed sensitivity from untested robustness. Across six market-model settings, Direct-minus-Unreg Sharpe ranges from −0.48 to +0.81; two period summaries show recovery from Direct while Unreg remains stronger. The evidence supports a conditional timing diagnosis, not calibrated execution performance, portfolio-scale robustness, or superiority over other architectures or constraint methods.

Keywords

portfolio reinforcement learning, regularization timing, transaction costs, risk control, financial data mining, non-stationary markets

CCS CONCEPTS

Computing methodologies ~ Machine learning ~ Learning paradigms ~ Reinforcement learning; Applied computing ~ Law, social and behavioral sciences ~ Economics.

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1 INTRODUCTION

Portfolio RL allocates assets under noisy returns, risk, and trading frictions. Online portfolio selection and direct RL established sequential allocation without explicit forecasting [1], [2].

Financial RL uses market representations and reusable environments [3], [4]. PPO, SAC, and TD3 are baselines [5]–[7]; cost-aware methods change rewards or impose constraints [8], [9].

Curriculum learning shows that signal order matters [10], [11]. We study when fixed cost-risk penalties enter training. Our exploratory timing analysis and training-only rule neither introduce a new architecture nor claim the first cost schedule. We distinguish optimizer seeds from market generalization.

2 RELATED WORK

Portfolio RL studies market representations and reproducible environments [2]–[4], [12]. Earlier work uses graph-based curricula [10] and a power-law cost ramp [13]. ACoDi's distinction is a turnover-triggered persistent switch, not delayed cost learning itself.

Automated curricula use progress or TD error [14], [15]; ACoDi has no teacher. MASTER [16] and FinRL-DeepSeek [17] change representations or signals; FinRL Contests [18] addresses evaluation.

CPO and RCPO use constraints or adaptive multipliers [8], [9]; their guarantees and safe-RL theory [19] do not apply to ACoDi's soft penalties. Drift motivates period tests [20]. Table 1 summarizes these distinctions.

PPO, SAC, and TD3 differ in learning rules [5]–[7], unlike encoder comparisons; MASTER [16] is not an RL baseline. Available ACoDi results cannot isolate architecture, reward, or constraint effects.

Portfolio RL and curriculum reviews [21], [22] frame reward-based TD3 [23] and hierarchical DDPG [24]. These alter learning or risk control rather than fixed-penalty timing.

Graph and Transformer encoders [25], [26] alter representation. Kruthof and Müller show friction-sensitive SAC evaluation [27]; these motivate validation but do not test ACoDi.

Table 1. Design comparisons; cited methods are not new experimental baselines.

Approach	Changed quantity	Control signal
ACoDi	Fixed penalty activation	Training turnover
PPO / SAC / TD3 [5]–[7]	Policy / critic update	Rollouts or value estimates
Cost ramp [13]	Cost level / reward target	Progress / future returns
Curriculum [15]	Task distribution	TD-error progress
MASTER [16]	Forecast representation	Market / stock features
CPO [8]	Constrained policy update	Estimated return and cost
RCPO [9]	Penalty multiplier	Constraint feedback

3 METHOD

3.1 Cost-Risk Timing

The policy observes past returns, weights, and risk/cost state, then selects nonnegative weights summing to one. Gross return uses next-period asset returns. PPO actor and critic are two-layer, 64-unit MLPs. Friction is a proportional-plus-quadratic turnover proxy; downside risk uses a trailing 60-step net-return history.

Let u denote the turnover quantity used by the shared cost implementation. At time t, the cost and net-return conventions are

$$C_t = 0.001u_t + 0.01u_t^2;\ R_t^{net} = R_t^{gross} - C_t. \quad (1)$$

At training checkpoint k, the activation weight a is in [0, 1], and the reported scheduled objective is

$$L_k = L_k^{PPO} + a_k(0.02L_k^{cost} + 0.05L_k^{risk}). \quad (2)$$

For a rollout segment B, the reported cost loss is the mean of its per-step costs. The risk loss is a CVaR-style estimate over the trailing window W, using losses equal to negative net returns:

$$L_B^{cost} = mean\{C_t : t \in B\}; \quad (3)$$

$$L_B^{risk} = CVaR_{0.05}(\{-R_i^{net} : i \in W\}). \quad (4)$$

The 60-step tail uses the largest 5% of losses; evaluation deducts the same friction proxy. A fixed historical penalty has zero current-policy gradient. Training needs a parameter-dependent estimator or reward/advantage pathway, neither specified; reward-level cost deduction is also unknown.

3.2 ACoDi Scheduler

Let T be the training budget; $f = t/T$ is elapsed fraction. At roughly 1,000-step probes, q_k is turnover and $\overline{q}_{recent}$ averages $h = 10$ probes. Initialize $a = 0$. Once $f \geq 0.4$, activate ($a = 1$) when early turnover exceeds 0.1 and recent turnover falls below 0.7 of it; otherwise activate at $f \geq 0.8$. This training-only switch persists. T, warm-up samples, and probe/update alignment remain unspecified.

Declining turnover is heuristic: a collapsed policy can also satisfy the trigger. Monitoring needs scalar summaries and short history, with no inference-time parameters. Extended settings switch at 20%, 50%, or 80%, or ramp over 40–80%; none identifies the separate main-table Delayed setting. Curriculum also changes fine-tuning learning rate. Multi-Task and Adaptive retain source labels because complete update rules are unavailable; they are not isolated architecture or reward ablations.

3.3 Persistent Activation and Computational Scope

Equation (5) defines the persistent switch. Let f_k be elapsed fraction, e_k early-reference turnover, and r_k recent turnover divided by e_k. Let H_k mark valid h-probe history and a positive reference. For notation, set $e_k = 0$ and $r_k = +\infty$ when $H_k = 0$. With $a_0 = 0$, the state after probe k is

$$a_k = max\{a_{k-1}, \mathbb{1}[f_k \geq 0.8], H_k\mathbb{1}[f_k \geq 0.4]\mathbb{1}[e_k > 0.1]\mathbb{1}[r_k < 0.7]\} \quad (5)$$

Here $\mathbb{1}$ is an indicator; ratios require $H_k = 1$. Activation persists after turnover rises; fallback begins at the first 80%-or-later probe. Monitoring costs $O(K)$ scalar work and $O(h)$ history, excluding turnover collection and policy execution; no runtime gain is measured.

4 EVALUATION PROTOCOL AND SCOPE

DS1 has ten assets: ZTS, A, AAPL, ABBV, ZBRA, ABT, ACGL, ACN, ADBE, and ADI. Daily adjusted-close training spans 2018-01-03 to 2023-08-08; testing spans 2023-08-09 to 2024-12-31. Features use past data, with training-fitted

normalization frozen at test and rolling features updated causally. Day-t state sets next-period weights. The data provider and corporate-action procedure are unspecified.

The shared PPO uses MLP [64, 64], learning rate 3×10^{-4}, 2,048-step rollouts, batch 64, 10 epochs, discount 0.99, GAE 0.95, clipping 0.2, and paired seeds 1, 7, 13, 21, 42, 50, 101, 123, 456, 777. Seeds sample optimization, not independent markets. Sharpe uses 252 days; MaxDD is positive.

Reference portfolios are equal weight, buy-and-hold, mean-variance, and risk parity; equal weight and risk parity reach Sharpe near 1.42, and a linear-cost PPO anchor reaches 3.07. Other summaries cover Nasdaq-100, sector ETFs, SAC/TD3, and market periods. Multiple inspected schedules make selected-stage results exploratory.

Tables and sweep values come from the original study; this revision adds no runs, data, or simulated observations. OpenAI Codex assisted the retrospective formalization of the activation rule (Eq. (5)), the cost-unit algebra (Eq. (6)), and Holm/Bonferroni calculations from reported p-values (Section 5.2); the authors checked these against the original summaries. Mean differences are descriptive. Pre-evaluation trigger settings lack a dated verification record. Selected rows and the sweep are exploratory because test-based schedule choice differs from training-only activation; missing run/window manifests prevent pooling panels.

5 RESULTS AND DISCUSSION

5.1 Reported PPO Benchmark

Table 2 summarizes the reported PPO means.

Table 2. Reported PPO means under the shared friction proxy.

Method	Sharpe	MaxDD %
Unreg PPO	4.13	4.96
Direct CR	3.79	5.22
Delayed CR	4.07	4.96
ACoDi	4.09	4.99
Multi-Task	4.06	4.97
Adaptive	4.00	4.99
Curriculum	4.22	4.86

Direct CR has lower Sharpe than Unreg; ACoDi recovers $0.297/0.341 = 87.1\%$ of the reported gap (88.2% from rounded table means). Curriculum's higher mean is not significant versus Unreg, and its changed learning rate precludes timing-only attribution. Unresolved cost units prevent a cost-savings claim (Section 5.6).

ACoDi improves displayed Sharpe by 0.30 and MaxDD by 0.23 points over Direct, but trails Unreg on both. No joint confidence region or execution-cost ranking is available.

5.2 Statistical Scope and Multiplicity

Table 3 applies retrospective Holm correction [28] to all seven reported paired comparisons. Sorted raw p-values are multiplied by $(8 - i)$ then cumulatively maximized and capped at one. For $p < 0.001$, adjusted < 0.007; remaining values are approximate from rounded inputs.

Table 3. Reported Sharpe differences and multiplicity adjustment.

Comparison	Δ	Raw p	Holm p
Direct – Unreg	−0.341	< 0.001	< 0.007
ACoDi – Direct	+0.297	< 0.001	< 0.007
ACoDi – Unreg	−0.044	0.042	0.126
Delayed – Direct	+0.282	< 0.001	< 0.007
Delayed – Unreg	−0.060	0.029	0.116
Multi-Task – Unreg	−0.066	0.049	0.126
Curriculum – Unreg	+0.088	0.353	0.353

Adjusted Direct–Unreg and both recovery comparisons remain below 0.05; deficits versus Unreg do not. Non-rejection is not equivalence. This reuses reported p-values; missing paired observations preclude reconstructed confidence intervals or market-level uncertainty.

Bonferroni across seven gives <0.007 for the three $p < 0.001$ pairs; adjusted values for ACoDi, Delayed, Multi-Task, and Curriculum versus Unreg are about 0.294, 0.203, 0.343, and 1.000. Across all 21 possible Table 2 pairs, cutoff $0.05/21$ (~0.00238) retains those three $p < 0.001$ results (<0.021 adjusted). Unreported pairs are not imputed. Both checks require valid source p-values and exclude other markets, metrics, and schedule searches.

5.3 Timing and Parameter Sensitivity

Fixed-switch starts at 0, 10, 20, 40, 60, 70, 80, 90% yield Sharpe 3.79, 3.74, 3.87, 3.85, 3.90, 3.93, 3.92, 3.93 (range 0.19; 70–90% span 0.01; no uncertainty bands). Separate Delayed (4.07) exceeds the sweep maximum (3.93), but no configuration, seed, or run mapping is available. These are not matched trials and cannot support a direct schedule comparison. This sweep does not validate ACoDi's trigger.

On fixed turnover/reference sequences and fallback, a higher threshold cannot delay first activation and later eligibility cannot advance it (Table 4). Retraining can change inputs and returns; the fixed-switch sweep does not test threshold or history length.

Table 4. Analytical parameter roles; no new training runs.

Parameter	Reported value	Fixed-input implication
Threshold	0.7	Higher cannot delay activation
History h	10 probes	No general monotone effect
Eligibility / fallback	0.4 / 0.8	Restrict candidates / force activation
Early-reference gate	> 0.1	Must scale with turnover units
Cost / risk weights	0.02 / 0.05	Inverse scaling preserves a loss term

Jointly rescaling recent and reference turnover preserves the relative test only if the absolute 0.1 gate is also rescaled; otherwise fallback may govern. History length has no monotone effect. A 60-step largest-5% risk tail uses only three losses, not enough for uncertainty or robustness claims.

For valid pre-fallback probes, a positive minimum distance between r_k and $\tau = 0.7$ preserves decisions under smaller threshold perturbations; equality can alter a strict test. Missing probe data prevent estimating this margin. Inverse

rescaling of a coefficient preserves its scaled loss term, so 0.02 and 0.05 cannot be compared without loss and gradient scales.

5.4 Markets and RL Algorithms

Table 5 presents the existing market-model comparisons.

Table 5. Existing exploratory market-model comparisons; Sharpe.

Setting	Unreg	Direct	Selected	Schedule
DS1 / SAC	2.94	3.75	3.75	Direct
DS1 / TD3	2.15	2.26	2.87	Ramp
N100 / PPO	3.09	2.90	3.09	Unreg
N100 / SAC	2.49	2.45	2.87	Ramp
N100 / TD3	2.45	2.09	2.89	Early
Sector / SAC	1.42	0.94	1.79	Ramp

Table 5 reports source-selected schedules, not validated optima: Ramp is 40–80%; Early lacks exact mapping. DS1/SAC favors Direct and N100/PPO Unreg. PPO, SAC [6], and TD3 [7] change learning rules; universes confound size and composition. These are boundary cases, not robustness tests.

Direct-minus-Unreg Sharpe across Table 5 is $+0.81$, $+0.11$ (DS1/SAC, DS1/TD3), and -0.19, -0.04, -0.36, -0.48 (N100/PPO, N100/SAC, N100/TD3, Sector/SAC): two positive, four negative. Selected-minus-Direct values are 0.00, 0.61, 0.19, 0.42, 0.80, 0.85, but Selected includes Direct and Unreg. Shared models and markets preclude treating rows as independent replications or estimating a population win rate.

5.5 Market Periods

Table 6 reports the existing market-period summaries.

Table 6. Existing regime diagnostics with drawdown magnitudes.

Method	Walk-forward Sharpe	High-vol Sharpe	High-vol MaxDD %
Unreg PPO	2.05	0.01	26.47
Direct CR	1.78	−0.28	32.27
Delayed CR	1.98	−0.05	28.36
ACoDi	1.97	−0.04	28.24
Multi-Task	1.99	0.01	27.18

ACoDi gains 0.19 walk-forward and 0.24 high-volatility Sharpe over Direct, recovering $0.19/0.27 = 70.4\%$ and $0.24/0.29 = 82.8\%$ of each Direct-to-Unreg gap. It trails Unreg by 0.08 and 0.05 Sharpe; high-volatility MaxDD is 1.77 points higher despite a 4.03-point reduction from Direct. Aggregate periods cannot establish reliability without dates, overlap, and retraining rules.

5.6 Cost Attribution and Execution Limits

Per-step L1 turnover between fully invested nonnegative weights is at most two, independent of asset count. Reported values exceed two, but units and aggregation are unknown; cost totals and the cost-performance scatter are therefore omitted. This bound covers neither cumulative turnover nor scale robustness.

A change of turnover units alone changes the apparent cost calibration. For a positive scale s, write $u' = s\,u$. Keeping the same numerical cost requires

$$C(u) = 0.001u + 0.01u^2 = (0.001/s)u' + (0.01/s^2)(u')^2 \quad (6)$$

Changing turnover units alters both proxy terms, quadratically for the second. Without units, coefficients are not calibrated fees or impact. Positive-domain derivative $0.001 + 0.02u$ confirms an increasing convex proxy, not execution realism. Spreads, volume, order size, and horizon are omitted; scenario Sharpe needs per-date costs and covariance with gross returns.

6 LIMITATIONS AND CONCLUSION

Reproducibility is limited by missing provider/adjustment, feature, execution, validation, and Sharpe details; N100/Sector constituents, window dates, budgets, and variant settings are incomplete. Completeness-based asset selection does not exclude survivorship bias.

Missing runs, independent market paths, original-test details, and gradient/cost-accounting pathways limit inference. The Delayed-sweep discrepancy remains unresolved; network, reward, constraint, portfolio-size, and retrained-parameter robustness are untested.

ACoDi recovers much of Direct's Sharpe loss in the reported PPO benchmark, yet market-model effects change sign and period summaries do not beat Unreg. Timing evidence, a formal persistent trigger, and analytical scale properties support conditional claims, not a universal schedule or calibrated deployment.

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